Overview

Walmart M5 data: 10 stores, 3 categories, Jan 2011 – May 2016

store_id: All

cat_name: All

Revenue, last 52 weeks
$45M

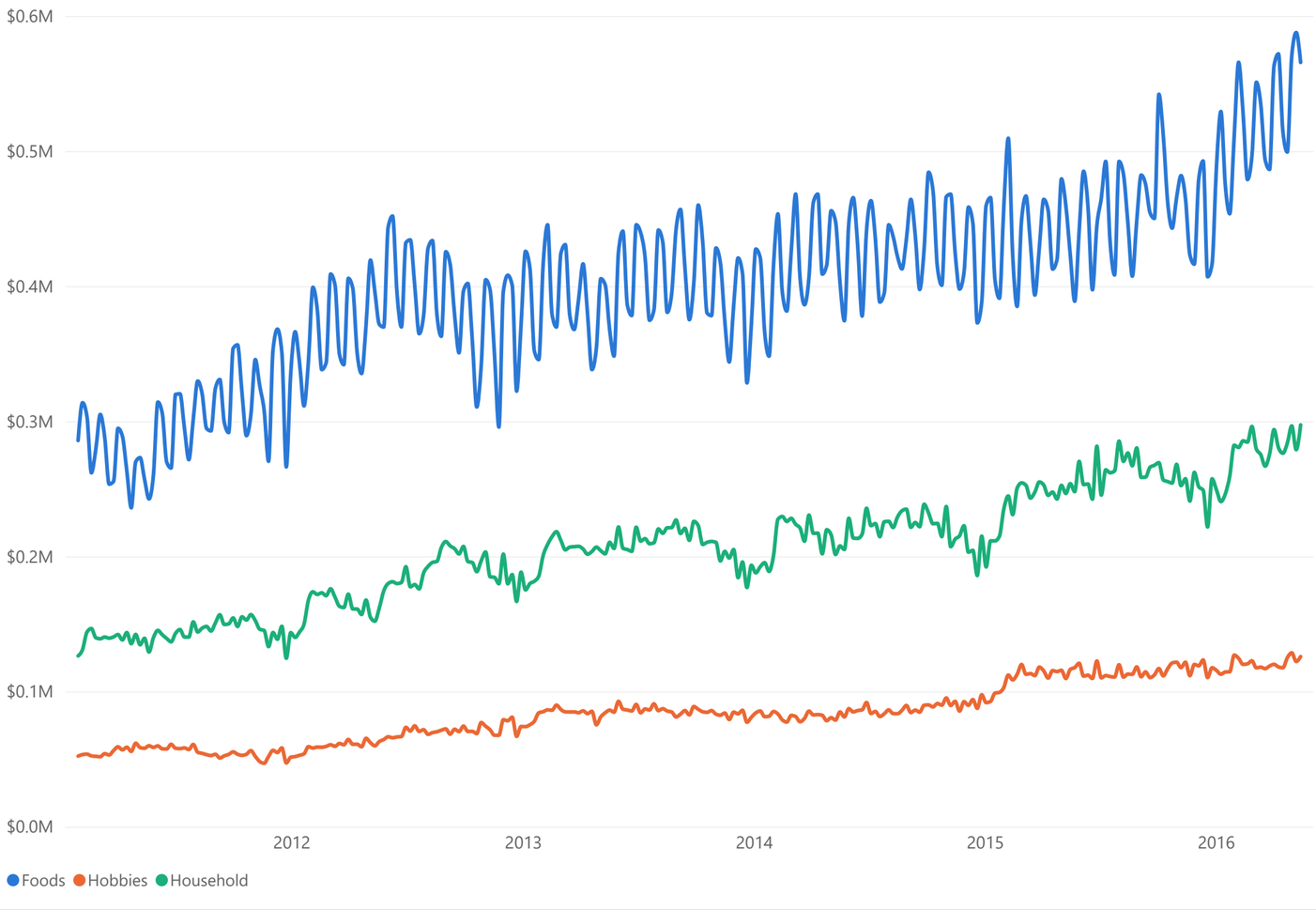
Lower forecast error vs baseline
32.2%

Estimated lost sales (stockouts)
$2M

Slow-moving item-stores
8K

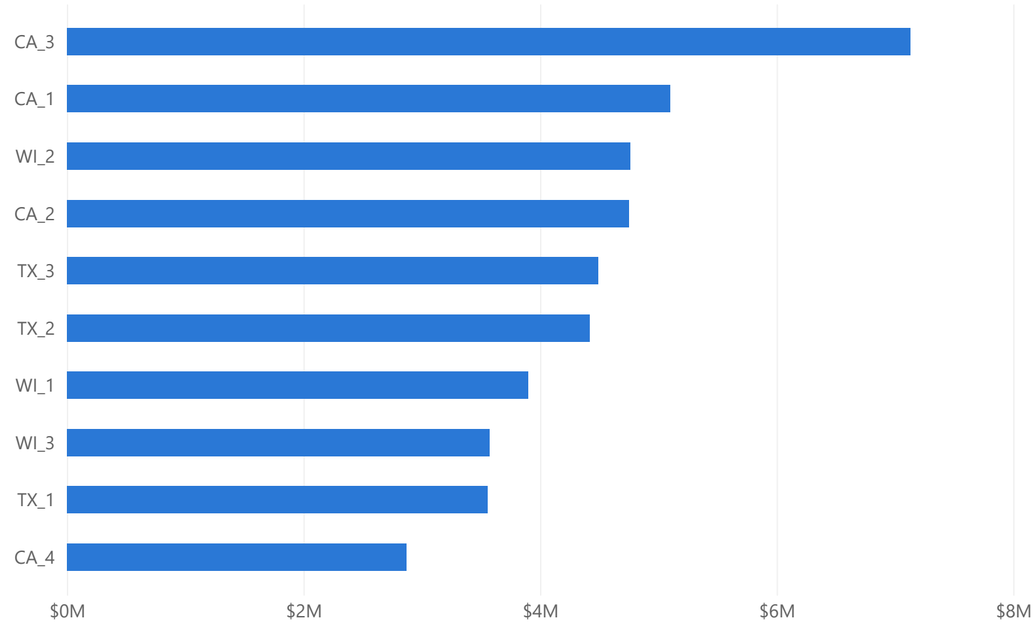
Revenue

by Date and cat_name



Revenue by store, last 52 weeks

Revenue 52w by store_id



- **LightGBM cuts forecast error 32% vs “same as last week” at category-store level.**
- **Likely stockouts cost an estimated $2.1M a year (28,970 events).**
- **All three categories are price-inelastic: shoppers respond only modestly to price changes.**

WAPE LightGBM
7.7%

WAPE Seasonal Naive
11.4%

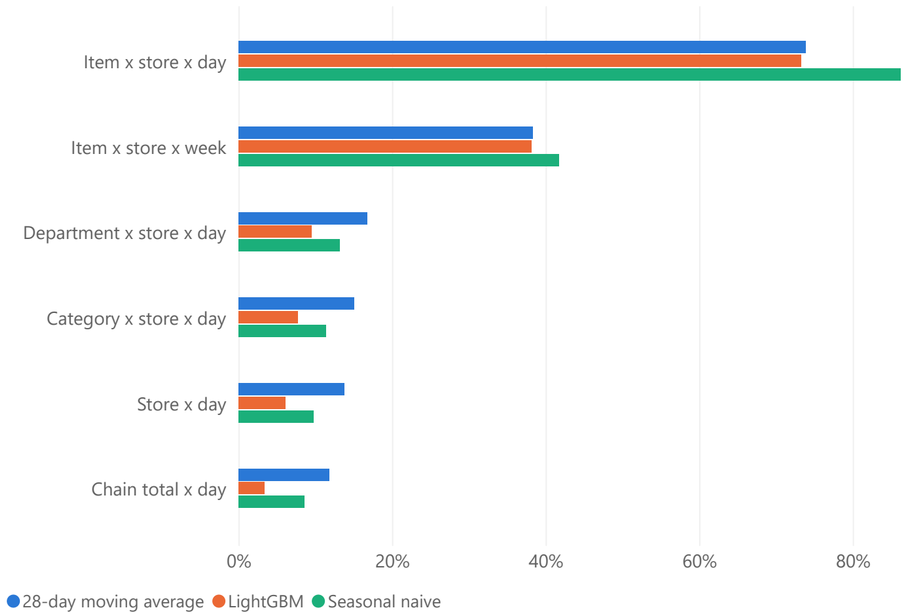
WAPE Moving Avg
15.1%

Forecast Bias
-2.0%

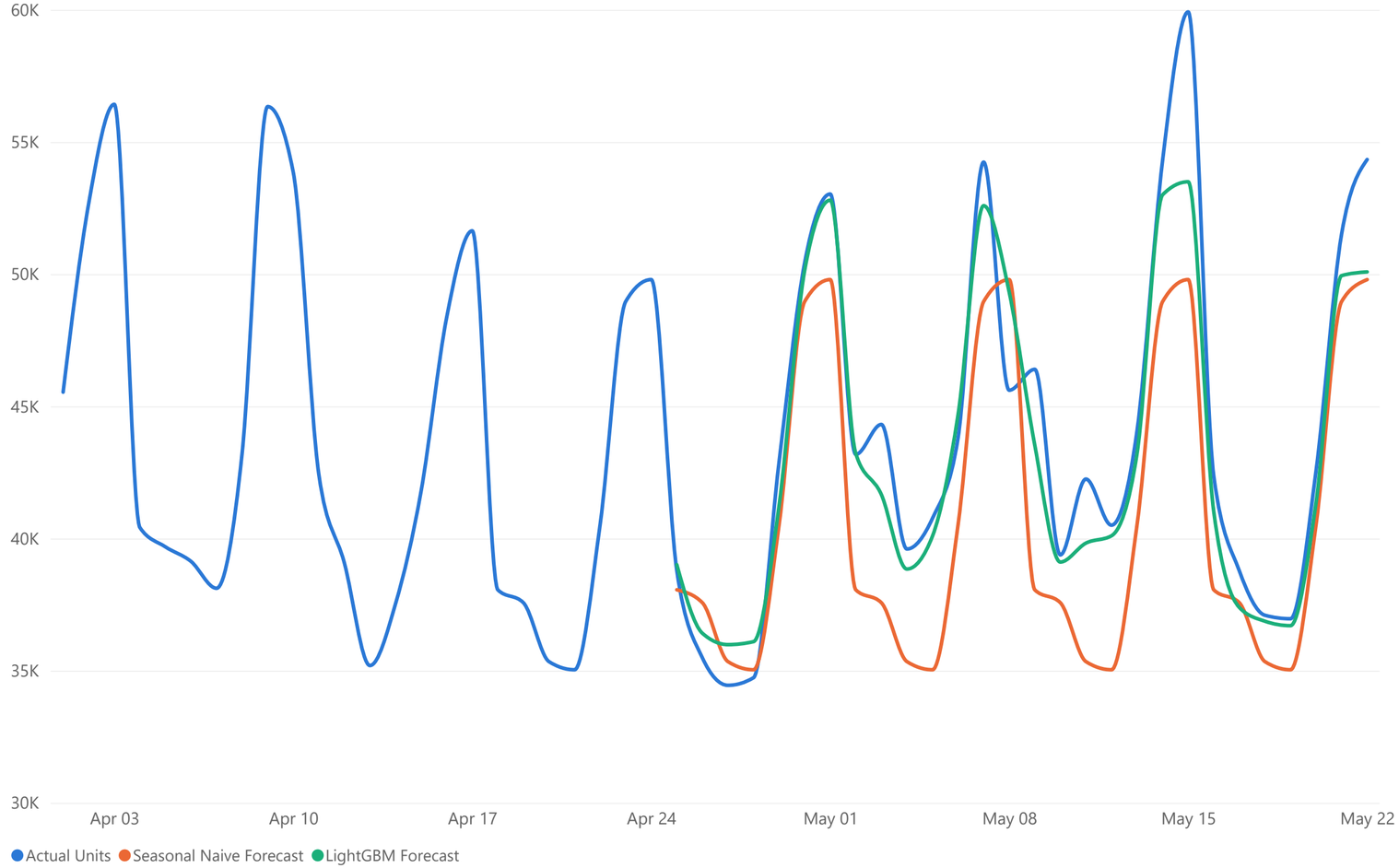
store_id
All

cat_name
All

Sum of wape
by level and model



Actual Units, Seasonal Naive Forecast and LightGBM Forecast
by Date



store_id	Foods	Hobbies	Household	**Total**
CA_1	4.7%	10.4%	7.0%	**5.8%**
CA_2	8.9%	12.2%	9.7%	**9.4%**
CA_3	8.1%	6.5%	4.3%	**6.9%**
CA_4	7.2%	14.7%	8.6%	**8.6%**
TX_1	7.2%	12.4%	10.1%	**8.4%**
TX_2	6.2%	13.3%	9.2%	**7.6%**
TX_3	7.4%	12.3%	7.2%	**7.8%**
WI_1	5.1%	9.3%	8.9%	**6.3%**
WI_2	8.7%	10.3%	9.9%	**9.0%**
WI_3	6.8%	13.3%	9.0%	**7.7%**
Total	**7.1%**	**11.2%**	**8.1%**	**7.7%**

What the forecast tells us
1. **32% more accurate where orders are planned.** Forecast error drops from 11.4% to 7.7% at category-store level, and from 8.6% to 3.4% on the chain-wide daily total.
2. **Less risk of running short.** The baseline under-forecasts by 7.4%; LightGBM by only 2.0%. Under-forecasting is what empties shelves.
3. **Better in 25 of 30 store-category combinations.** Hobbies is hardest to predict (11.2% error vs 7.1% for Foods); CA_4 Hobbies is the weakest spot (14.7%).
4. **Plan at category and store level.** For one item on one day, the model is barely better than a 28-day average, so item orders still need safety stock.

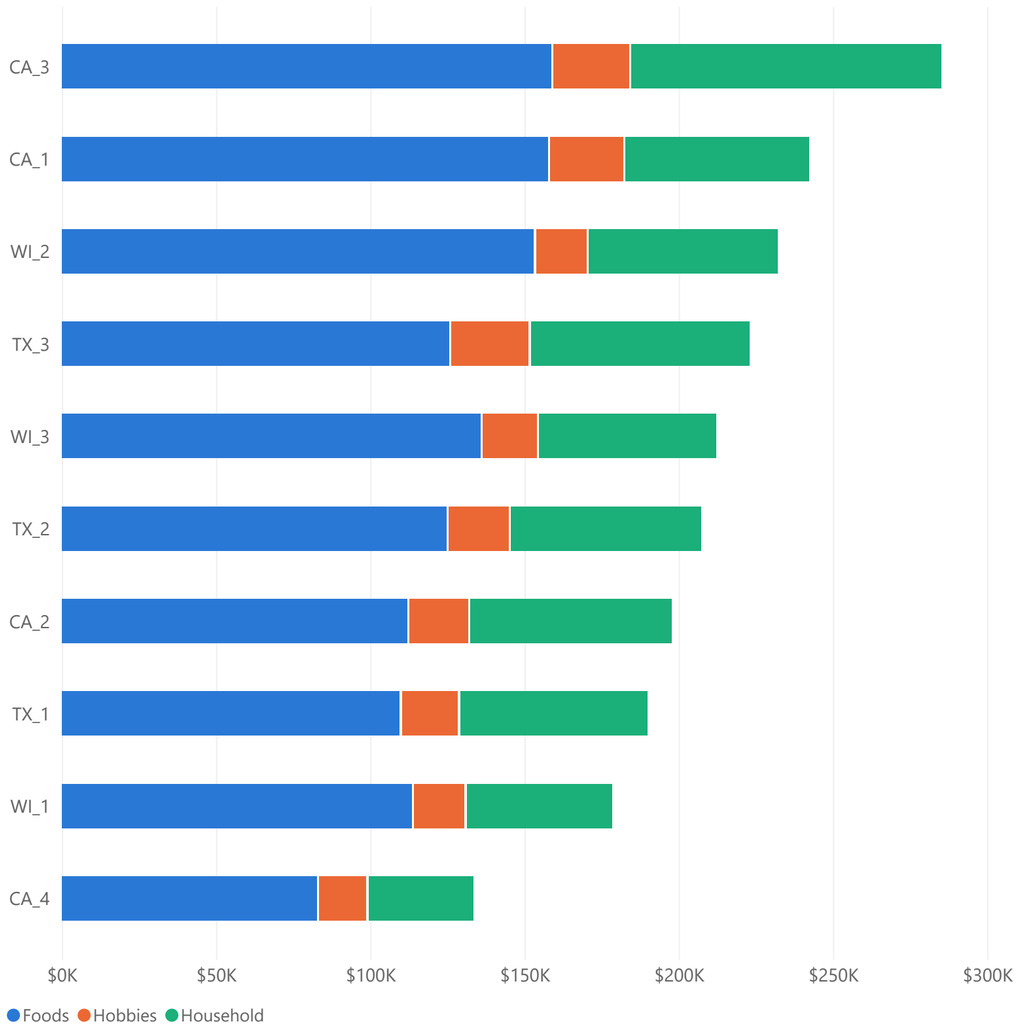
Est Lost Revenue
$2M

Stockout Events
29K

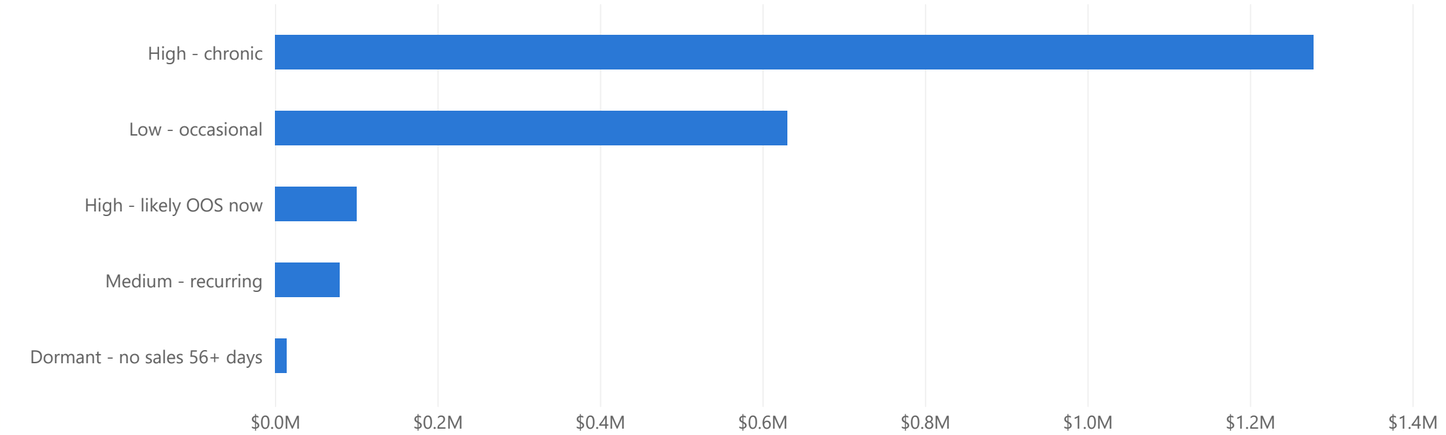
Chronic Item-Stores
7K

Likely Out of Stock Now
483

Est Lost Revenue
by store_id and cat_name



Est Lost Revenue
by risk_tier



item_id	store_id	risk_tier	Sum of stockout_days_365d	Est Lost Revenue
FOODS_3_202	TX_3	High - chronic	60	$9,592
FOODS_3_202	TX_1	High - chronic	60	$7,470
FOODS_3_202	TX_2	High - chronic	59	$7,413
FOODS_3_090	TX_3	High - chronic	65	$6,975
FOODS_3_090	CA_1	High - chronic	64	$6,734
FOODS_3_090	WI_3	High - chronic	65	$6,616
FOODS_3_090	TX_2	High - chronic	65	$6,582
FOODS_3_202	WI_3	High - chronic	59	$6,353
FOODS_3_202	CA_1	High - chronic	60	$6,128
FOODS_3_090	TX_1	High - chronic	65	$5,478
FOODS_1_096	WI_3	High - chronic	56	$4,106
FOODS_3_785	CA_1	High - chronic	56	$3,978
FOODS_3_090	CA_2	High - chronic	65	$3,914
FOODS_3_120	TX_2	High - chronic	33	$3,901
FOODS_3_120	CA_1	Low - occasional	16	$3,742

Where empty shelves cost the most

1. A small group drives most of the loss. 7,019 chronic item-stores account for 60% of the estimated $2.1M; the worst 10% of affected item-stores account for 48%.

2. Foods carries 60% of the loss ($1.27M), Household 30%, Hobbies 10%.

3. Two products repeat across stores. FOODS_3_090 and FOODS_3_202 make up 11 of the top 15 rows, across 6 stores. That points to a supply or ordering issue rather than one store's execution.

4. 483 item-stores were likely empty at the end of the data: the first list to check

Sell Rate

40.9%

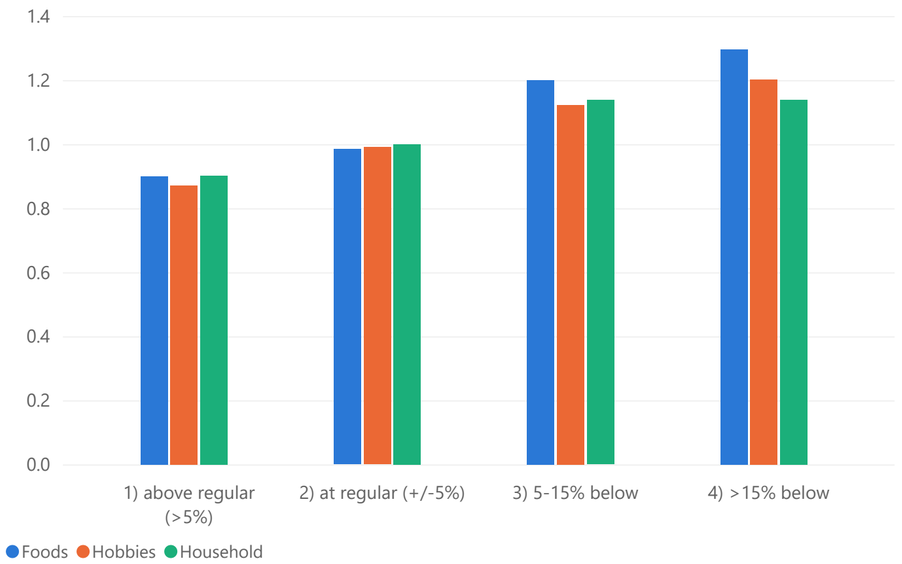
Slow Movers

8K

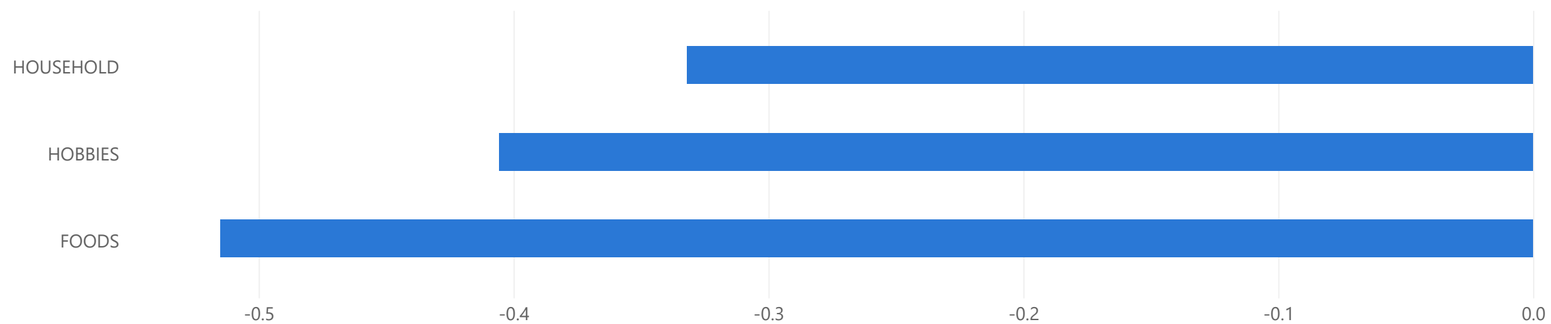
Sell rate shows how often an item sells on the days it's on the shelf. Low sell rate = slow-moving stock that ties up space and cash.
Elasticity shows how much volume changes when price changes: −0.5 means a 10% price cut lifts units about 5%.

dept_id	CA_1	CA_2	CA_3	CA_4	TX_1	TX_2	TX_3	WI_1	WI_2	WI_3	**Total**
FOODS_1	48.6%	55.4%	53.4%	39.4%	36.8%	40.5%	38.1%	49.6%	42.9%	39.5%	**44.4%**
FOODS_2	46.3%	47.0%	53.6%	40.5%	39.1%	41.5%	45.2%	54.8%	55.6%	43.2%	**46.7%**
FOODS_3	55.5%	53.2%	61.9%	45.4%	42.5%	48.7%	45.8%	53.9%	51.8%	47.6%	**50.6%**
HOBBIES_1	40.6%	35.3%	43.0%	33.0%	26.9%	34.2%	32.0%	36.4%	26.4%	26.1%	**33.4%**
HOBBIES_2	18.2%	19.2%	22.5%	9.2%	15.5%	20.7%	19.1%	15.9%	13.2%	13.7%	**16.7%**
HOUSEHOLD_1	53.9%	53.2%	60.4%	39.6%	49.2%	51.7%	50.3%	47.4%	50.6%	42.7%	**49.9%**
HOUSEHOLD_2	26.2%	31.6%	32.9%	17.7%	21.1%	24.9%	21.5%	21.4%	17.1%	18.2%	**23.3%**

Sum of avg_units_index
by price_band and cat_name



Price Elasticity
by segment



What sells, what sits, and what price does
1. Foods sells most consistently (48.6% of listed days), Hobbies least (29.0%). HOBBIES_2 is the slowest department in all 10 stores.
2. 7,770 slow movers (listed 180+ days, sold on fewer than 20% of them). Household has the most (3,438): the first assortment to review.
3. Shoppers are price-inelastic. A 10% price change moves volume about 3–5%. Discounts deeper than 15% go with 1.30x normal weekly units in Foods, but in Household they add almost nothing over a 5–15% discount (1.14x either way)